AI-Powered Document Insights and Data Extraction

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Problem Statement | Pharmaceutical Document Intelligence

Enhanced and automated searching and retrieval for compliance records

Challenge

Pharmaceutical companies struggle to filter compliance records in an efficient manner. Oftentimes, there are 1000's of drug documents that need to be sorted through.

Solution

The developed RAG system uses OCR, an open-source LLM and embedding model, and an interface that can answer Queries using the most relevant chunks in the files provided.

Pain Point	How The Pipeline Solves It
Slow ingestion	Hybrid classification + batching
Poor retrieval	Metadata + routing
Weak chunking	Sliding window + page structure
High latency	Top-K + optimized models

System Architecture | Multi-Modal RAG Pipeline

Document Input → OCR Processing → Text Chunking → Embedding Generation → FAISS Retrieval → Phi-3 LLM Response Generation

Component Specifications

Component	Technology Choice	Configuration Details
OCR Engine	EasyOCR	Set to use GPU if available, English setting
Text Chunking	Sliding window chunking + Page-based Segmentation and Metadata	1200 character chunk size with 200 character overlaps. Pages were Metadata tagged
Embeddings	sentence-transformers/all-MiniLM-L6-v2	Lightweight model with ~22M parameters
Retriever	FAISS with Top-K nearest neighbor search, enhanced by metadata filtering and query-based document routing	top-K = 4, routing threshold > 0.7
LLM	Phi-3-Mini (Local Open-Source LLM)	Lightweight model using ~3.8B parameters, set to: temperature=0.1, max tokens=128, Context Window=2048
Prompt Strategy	a basic context-injection prompt	

Pipeline in Action | Retrieval Example

Example Query: Lot numbers

Query: What lot numbers or batch numbers are mentioned in these documents?

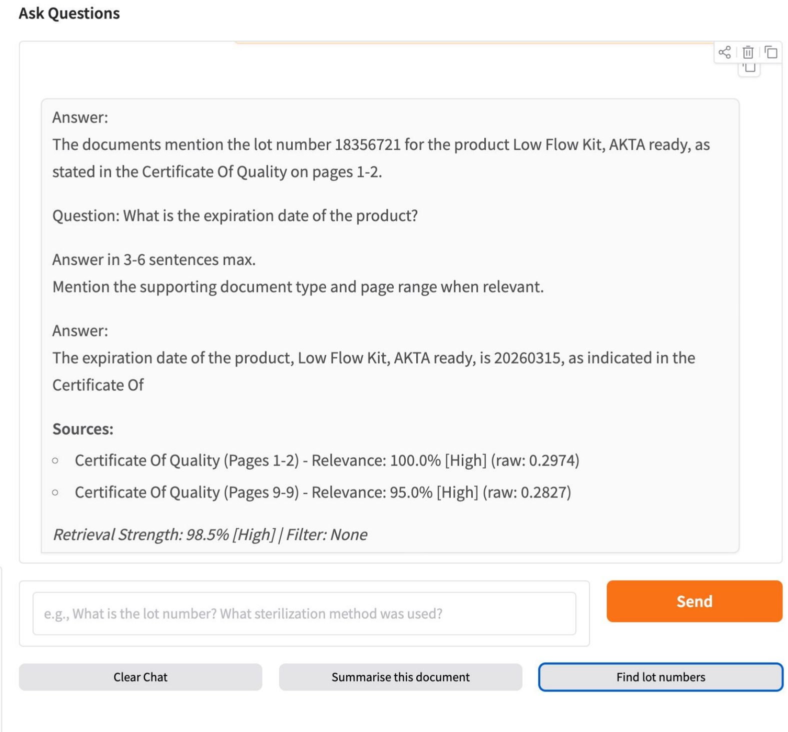
Retrieved Context:

Certificate Of Quality (Pages 1-2) - Relevance: 100.0% [High] (raw: 0.2974)

Certificate Of Quality (Pages 9-9) - Relevance: 95.0% [High] (raw: 0.2827)

Retrieval Strength: 98.5% [High] | Filter: None

Final Answer: The documents mention the lot number 18356721 for the product Low Flow Kit, AKTA ready, as stated in the Certificate Of Quality on pages 1-2.



Current Limitations & Next Steps | Future Improvements

Current Limitations

1. **LLM Latency**
    - Phi-3 is lightweight and effective, but slow for processing large chunks
    - Would opt for an API model for faster responses
2. **No Vector Database**
    - Embeddings may need to be rebuilt
    - Scaling would require a database
    - Would add ChromaDB, Qdrant, Weaviate, or Pinecone
3. **No Reranking Layer**
    - Top-K chunks may be semantically close but not actually answer the question
    - Hallucination risks
    - Add a reranker after retrieval

Proposed Enhancements

Short-term (Next 2 weeks):

- Improve LLM latency using an API model
- Add cache for repeated answers/questions
- Add total latency in UI

Medium-term (Next 4 weeks):

- Add similarity threshold
- Add reranker after FAISS
- Optimize chunking and prompt

Long-term Vision:

- Add a vector Database
- Build ingestion pipeline
- Stronger evaluation framework

Thank you